NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Week 6 – Supply of Maritime Transport





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17 September 2026

Learning Objectives

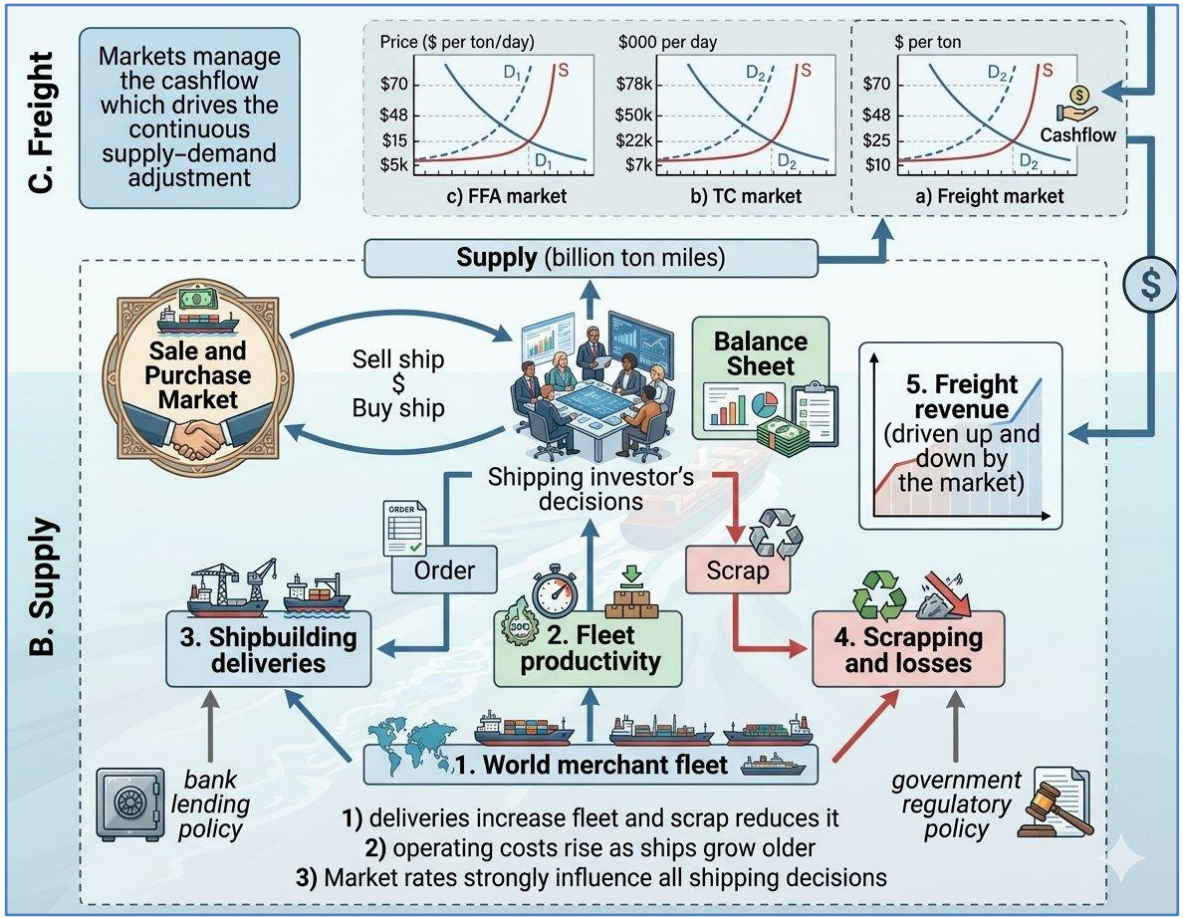
1. Explain that the supply of shipping services is very inelastic.
2. Identify the decision-makers controlling the supply.
3. Discuss the trends in the world merchant fleet composition.
4. Elaborate the factors that contribute to fleet productivity.

Learning Objectives

5. Analyse the shipbuilding production for each sector.
6. Analyse the factors influencing scrapping and losses.
7. Explain how shipowners can increase or decrease their supply in the short-run and long-run in response to changing freight rates.

1. Introduction

1. Introduction



1. Introduction

- Supply of shipping services is slow and ponderous in its responses to changes in demand.
- Ships generally take at least a year to build and delivery may take 2-3 years under the peak market cycle.
- Prevents the market from responding promptly to any sudden upsurge in demand.
- Once built, the ships have a 20-30 years lifespan, so responding to a fall in demand is a lengthy business, particularly when there is a large surplus to be removed.

1.1 Supply of Shipping Services

LO1

The term supply refers to a **functional relationship between freight rates and the quantity supplied by carriers**.

The supply of sea transport is strongly influenced by freight rates.

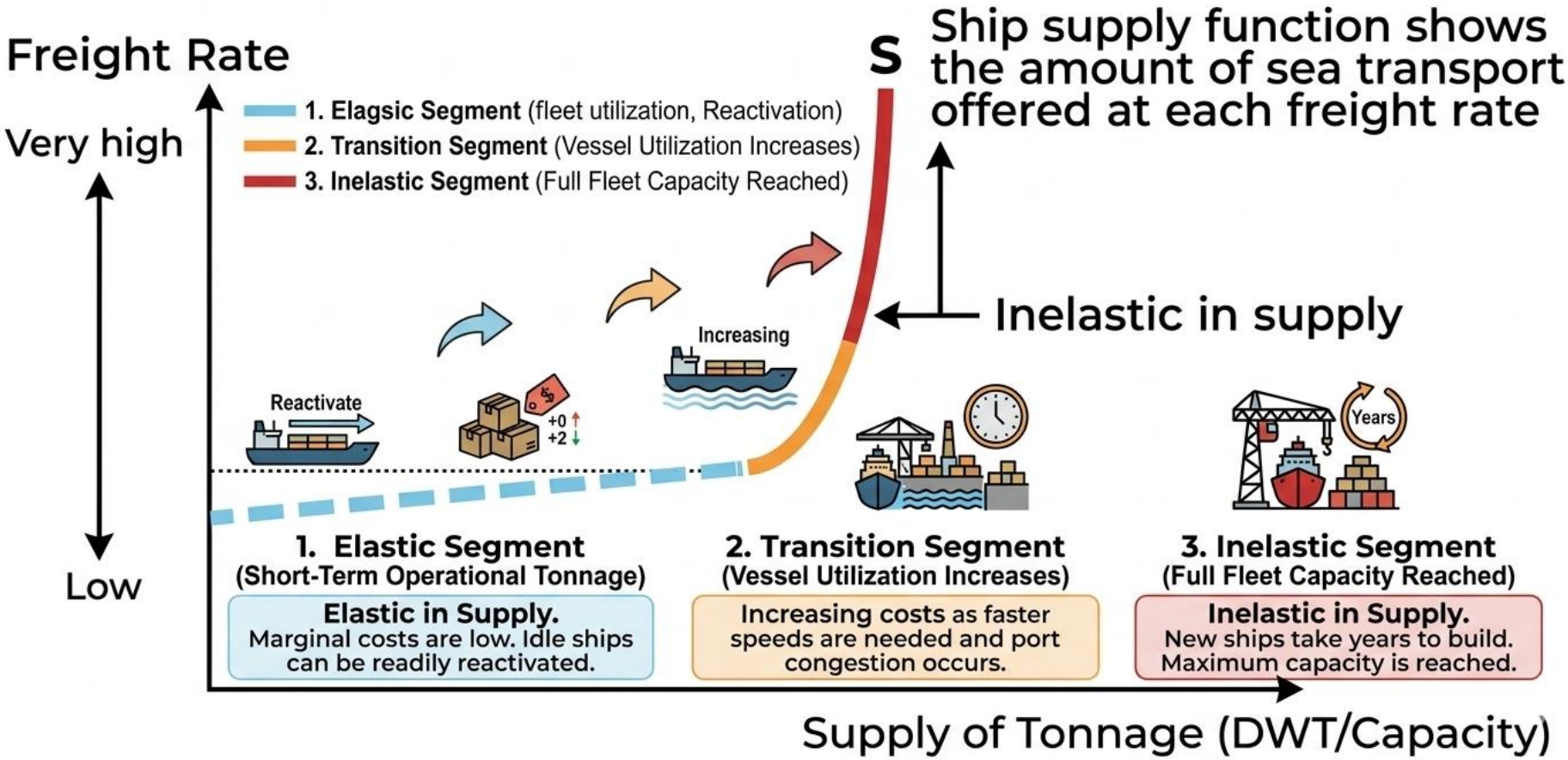
The shipping supply function shows the in and out of ships in response to freight rates.

If freight rates fall below operating costs, ships go to lay-up and supply is reduced.

1.1 Supply of Shipping Services

The slope of the shipping supply curve depends on three factors:		
Bigger ships have lower transport costs per unit of cargo. Hence, _____ **ships will have a _____ lay-up point**. This drives the smaller or inefficient ships into lay-up during recessions.	**_____ ships have _____ operating costs so the lay-up point will occur at a _____ freight rate**.	**When all the available tonnages are in use, supply of tonnage can only be increased with higher speeds and improvement in operational efficiency of ships**. Under such circumstances, there will be a steep slope in the shipping supply curve.

1.1 Supply of Shipping Services



1.1 Supply of Shipping Services

- Price elasticity of shipping supply measures its responsiveness to changes in freight rates.
- During recessions, the supply of sea transport tends to be very elastic when many ships are laid up.
- When all ships are in service, the supply becomes _____.

1.1 Supply of Shipping Services

Supply of sea transport is measured in terms of the supply of tonnage, which refers to the **available capacity for carrying cargo** from one or more ports to one or more ports by sea.

All ships that are trading in the freight market constitute to '_____ **shipping supply**'.	Ships that are not trading , i.e., laid-up tonnage, constitute '_____ **shipping supply**'.	All ships that are suitable for trading, i.e., active shipping supply, and available shipping supply constitute the **total shipping supply**.



1.2 Decision-Makers

LO2

The supply of ships is affected by **four** parties.

_____ decide whether to order new ships or scrap old ships.	_____ influence shipowners by booking shipping space to transport their cargoes.	_____ influence investment as lenders to finance ship purchase.	_____ affect fleet capacity through safety or environmental legislations.



2. World Fleet

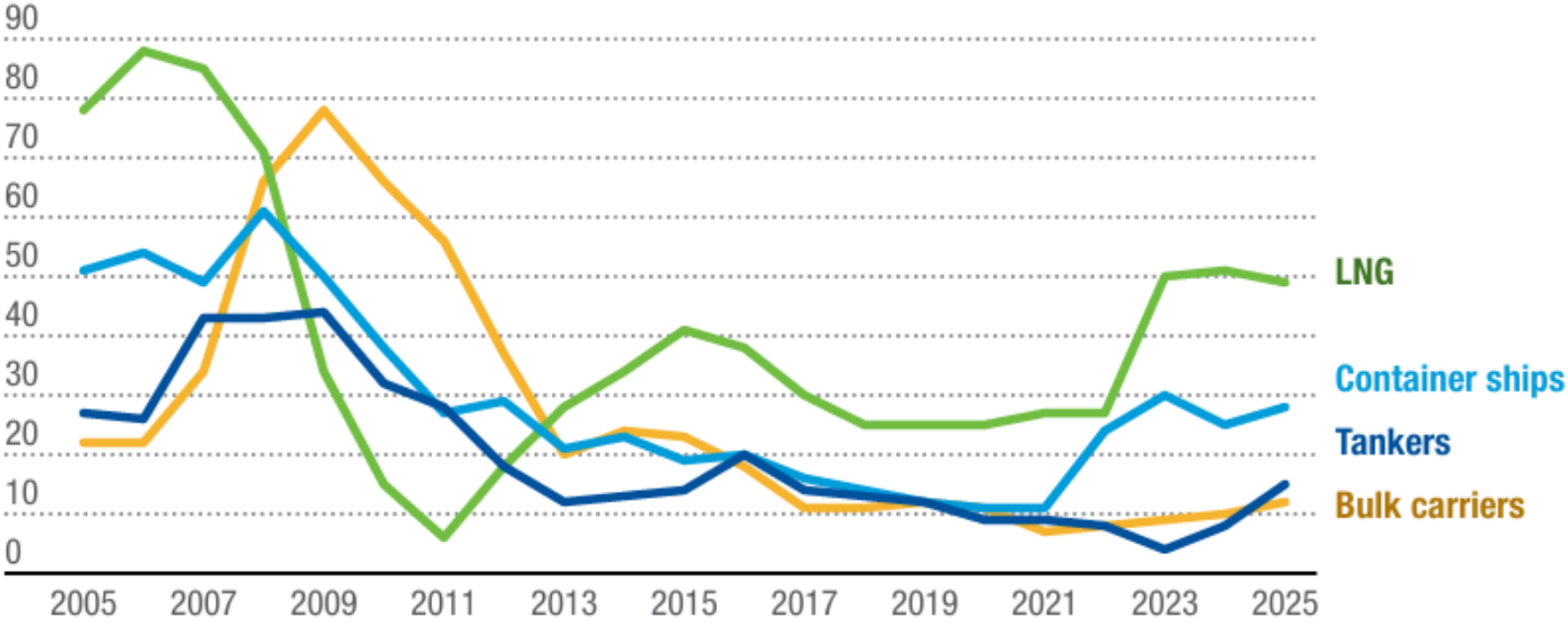
2. World Fleet

- In the long run, deliveries of new ships and scrapping of old ships determine the rate of fleet growth.
- Since average life span of ship is 25 years, only a small proportion of the fleet is scrapped each year.
- Key feature of shipping – how supply adjusts, when ship demand does not turn out as expected.

Table II.5
Trends in world fleet capacity by vessel type, 1 January 2025

Fleet composition by vessel type	Indicator	2024	2025	Change (percentage)
Bulk carriers	Dead weight tons (thousands)	1 006 329	1 036 158	3.0
	Share (percentage)	42.6	42.5	
Oil tankers	Dead weight tons (thousands)	665 472	669 842	0.7
	Share (percentage)	28.2	27.5	
Container ships	Dead weight tons (thousands)	331 567.4	363 759.1	9.7
	Share (percentage)	14.0	14.9	
Other types of ships	Dead weight tons (thousands)	271 141	282 651	4.2
	Share (percentage)	11.5	11.6	
Offshore supply	Dead weight tons (thousands)	89 507	91 334	2.0
	Share (percentage)	3.8	3.7	
Liquefied gas carriers	Dead weight tons (thousands)	93 907	100 462	7.0
	Share (percentage)	4.0	4.1	
Chemical tankers	Dead weight tons (thousands)	52 584	54 333	3.3
	Share (percentage)	2.2	2.2	
Other/n.a.	Dead weight tons (thousands)	26 326	27 576	4.7
	Share (percentage)	1.1	1.1	
Ferries and passenger ships	Dead weight tons (thousands)	8 818	8 947	1.5
	Share (percentage)	0.4	0.4	
General cargo	Dead weight tons (thousands)	85 809	87 421	1.9
	Share (percentage)	3.6	3.6	
World total	Dead weight tons (thousands)	**2 360 319**	**2 439 832**	3.4

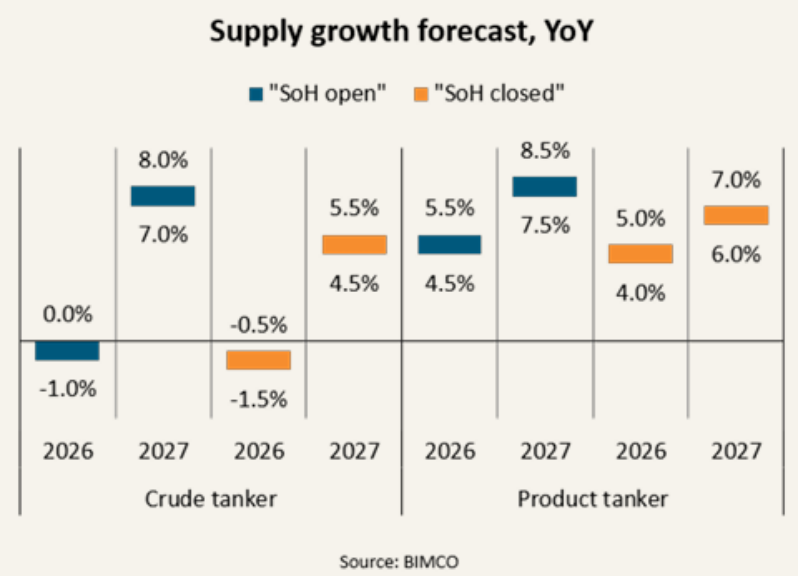
Figure II.10
Types of ships ordered
(Percentage of global active fleet gross tonnage)

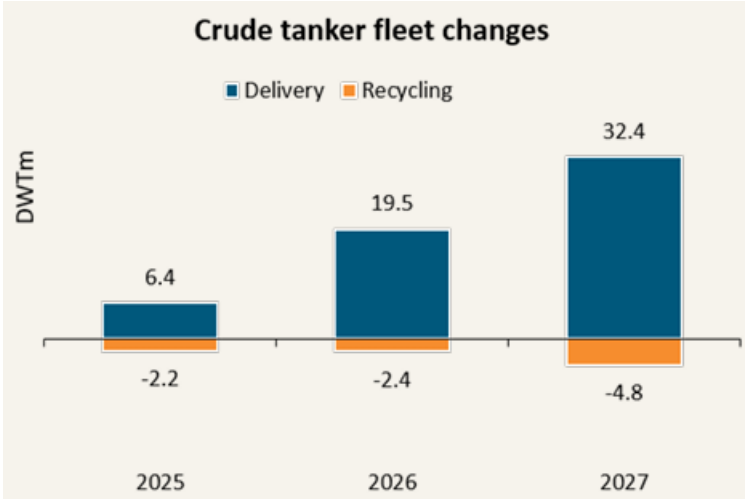


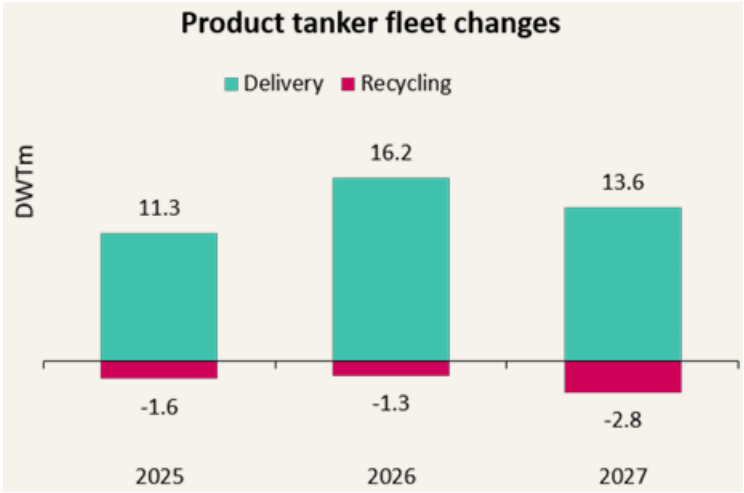
Container Market

Rank	Operator	Total		Owned		Chartered			Orderbook		
		Teu	Ships	TEU	Ship...	TEU	Ships	% Chart	TEU	Ship...	% existing
1	Mediterranean Shg Co	7,424,319	1013	4,763,130	763	2,661,189	250	35.8%	2,919,618	165	39.3%
2	Maersk	4,745,587	752	2,943,996	359	1,801,591	393	38%	1,211,002	94	25.5%
3	CMA CGM Group	4,402,124	724	2,919,095	381	1,483,029	343	33.7%	1,740,301	163	39.5%
4	COSCO Group	3,650,609	566	2,202,330	210	1,448,279	356	39.7%	1,912,546	155	52.4%
5	Hapag-Lloyd	2,396,281	286	1,455,918	134	940,363	152	39.2%	499,352	64	20.8%
6	ONE (Ocean Network Express)	2,179,882	271	1,037,548	111	1,142,334	160	52.4%	584,268	47	26.8%
7	Evergreen Line	2,022,899	242	1,479,527	169	543,372	73	26.9%	868,460	76	42.9%
8	HMM Co Ltd	1,036,129	98	847,404	72	188,725	26	18.2%	220,642	36	21.3%
9	Yang Ming Marine Transport Corp.	755,680	98	382,296	63	373,384	35	49.4%	272,260	21	36%
10	Zim	700,052	115	92,934	15	607,118	100	86.7%	181,836	23	26%
11	Wan Hai Lines	624,354	124	624,354	124				504,510	49	80.8%
12	PIL (Pacific Int. Line)	472,200	102	334,825	83	137,375	19	29.1%	305,798	29	64.8%
13	X-Press Feeders Group	186,652	106	109,624	56	77,028	50	41.3%	65,400	13	35%
14	SITC	186,241	120	163,687	99	22,554	21	12.1%	66,232	37	35.6%
15	KMTC	157,748	66	86,021	29	71,727	37	45.5%	82,056	12	52%
16	Global Feeder Shipping LLC	148,569	58	59,977	25	88,592	33	59.6%	7,600	2	5.1%
17	DP World	147,680	85	13,582	2	134,098	83	90.8%	74,161	11	50.2%
18	IRISL Group	140,184	27	140,184	27						
19	TS Lines	125,784	45	106,059	38	19,725	7	15.7%	69,432	15	55.2%
20	Sinokor Merchant Marine	124,496	68	122,711	67	1,785	1	1.4%	52,000	4	41.8%

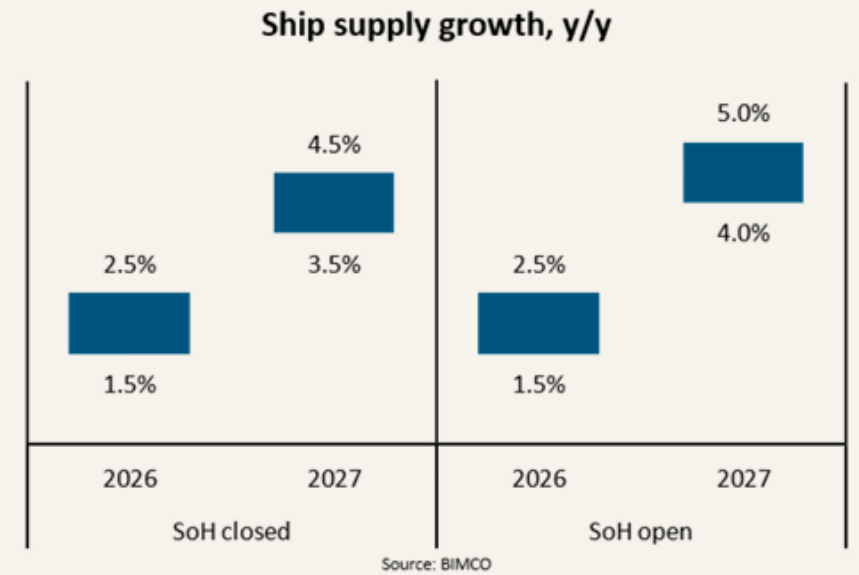
Tanker Market

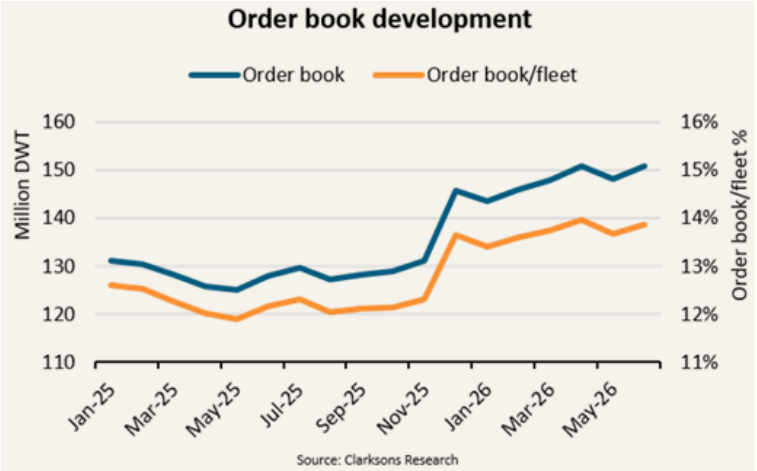


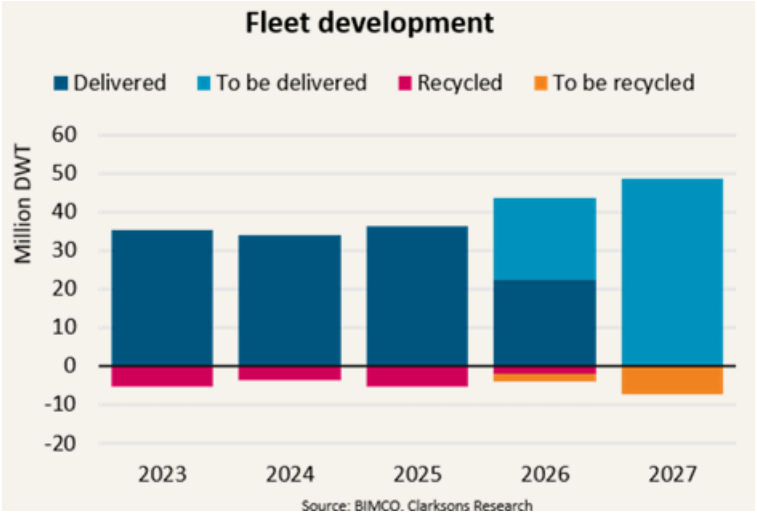




Dry Bulk Market







Review Question

- When all ships are in service, supply is said to be inelastic.

a. True

b. False

Answer:

Review Question

- Which of the following does not influence the supply of maritime transport services?

a. Bankers

b. Regulatory authorities

c. Shipowners

d. Ship agents

Answer:



3. Fleet Productivity

3. Fleet Productivity

- Measure in ton-miles per deadweight.

$$\frac{\text{quantity of cargoes shipped (ton) x distance shipped (miles)}}{\text{total deadweight tonnage}}$$

- Four factors influencing productivity:
  a. **Speed**
  b. **Port time**
  c. **Deadweight utilisation**
  d. **Loaded days at sea**

3.1 Speed

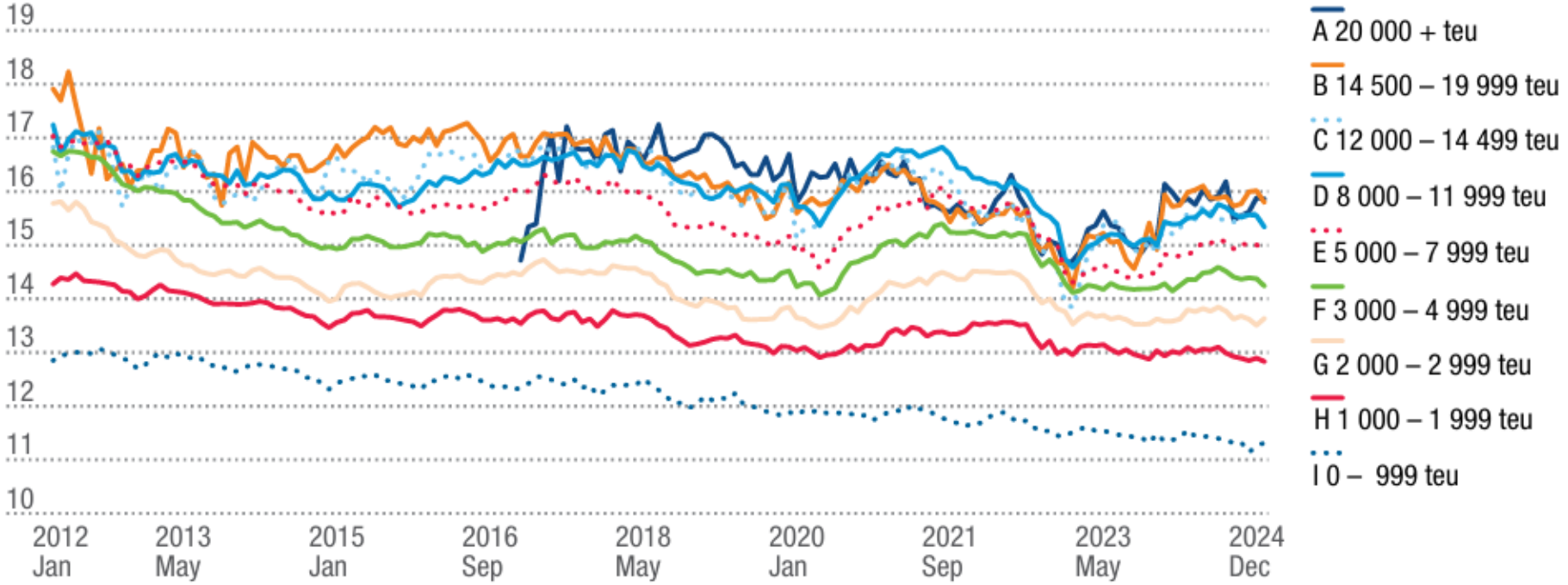
- Refers to the _____ _____.
- When faced with depressed freight market and improving cargo efficiency (ton-miles) is not possible, shipowners reduce their operational costs or remove vessels from trading.
- To save bunker fuel costs, owners _____.
- Eventually, ships too expensive to operate are laid up.

3.1 Speed

Figure II.14

Monthly average speed of container vessels

(Knots)



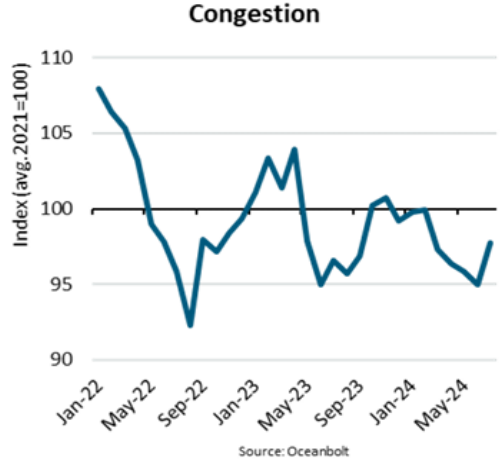
3.2 Port Time

Port _____
(increase demand, limited port capacity, pandemic, extreme weather, accidents)

Port _____

Additional burden to carriers & shippers

Lose business competitiveness



We expect that a reduction in **congestion** could lead to a 0-1% y/y increase in supply in 2024. Ship capacity in congestion fell primarily in the capesize and panamax segments, in Australia and Brazil respectively.

3.3 Deadweight Utilisation

- Refers to the _____ *been utilised to carry cargo*.
- Remaining capacity not utilised or lost due to bunkers, stores or less than full-load quantities stipulated in the charter-party.
- During recessions/random shocks, deadweight utilisation reduced as there is increasing tendency for owners to carry part cargoes to cover operating costs or reposition vessels.

3.4 Loaded Days at Sea

- A vessel's time is divided between loaded days at sea and '_____' days (in ballast, port or off hire).
- Reduction in unproductive time allows an increase in loaded days at sea.
- Vessels designed for cargo flexibility can improve their loaded time at sea because they are able to switch their cargoes for backhaul voyages.

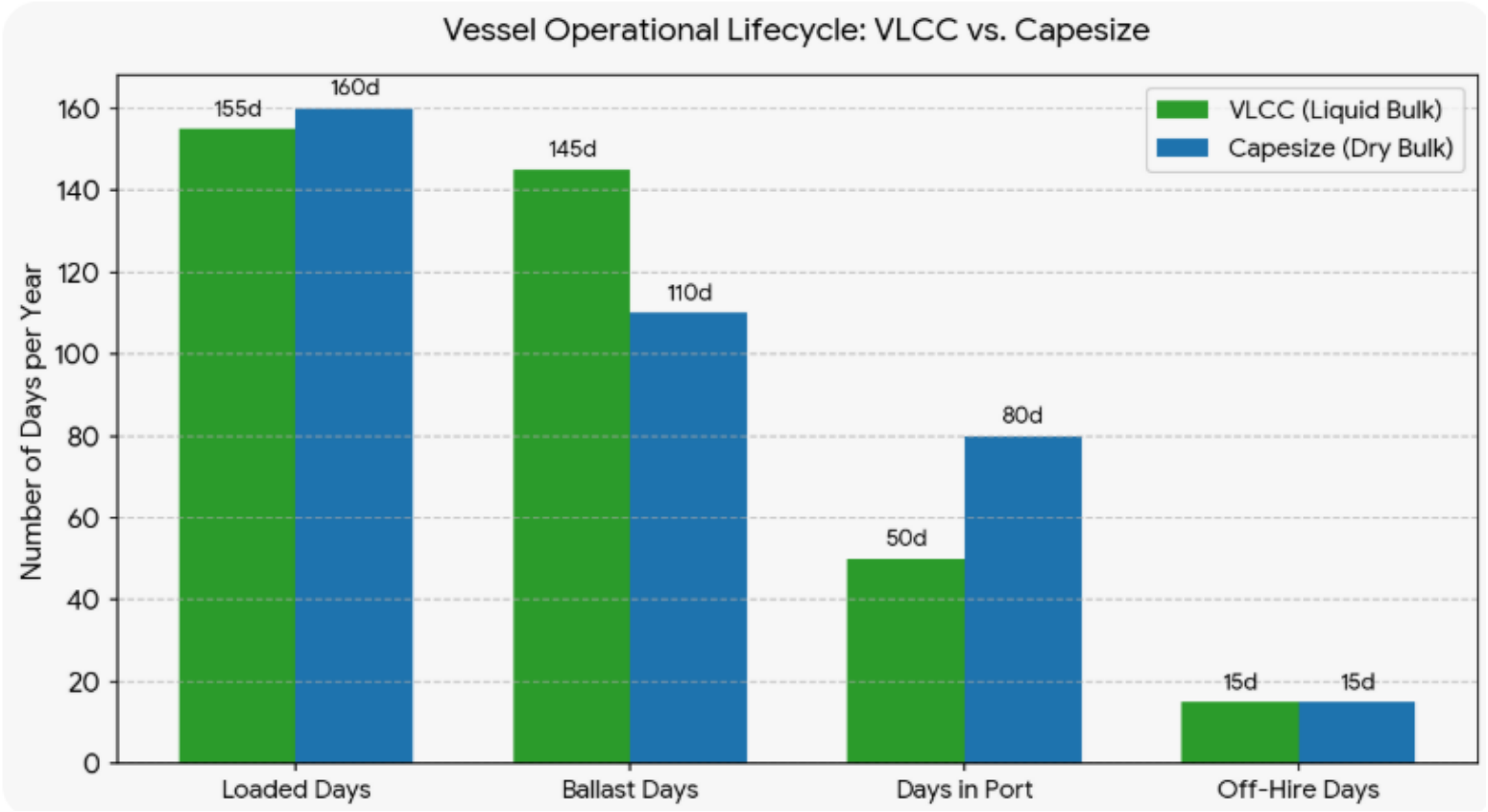
Combination Carriers



COMBINATION CARRIER VESSEL BARU (PICTURED) IS IN KLAVENESS COMBINATION CARRIERS' CLEANBU FLEET. IT CARRIES CLEAN PETROLEUM PRODUCTS AS WELL AS DRY BULK COMMODITIES.

Source: Klaveness

3.4 Loaded Days at Sea



LO5

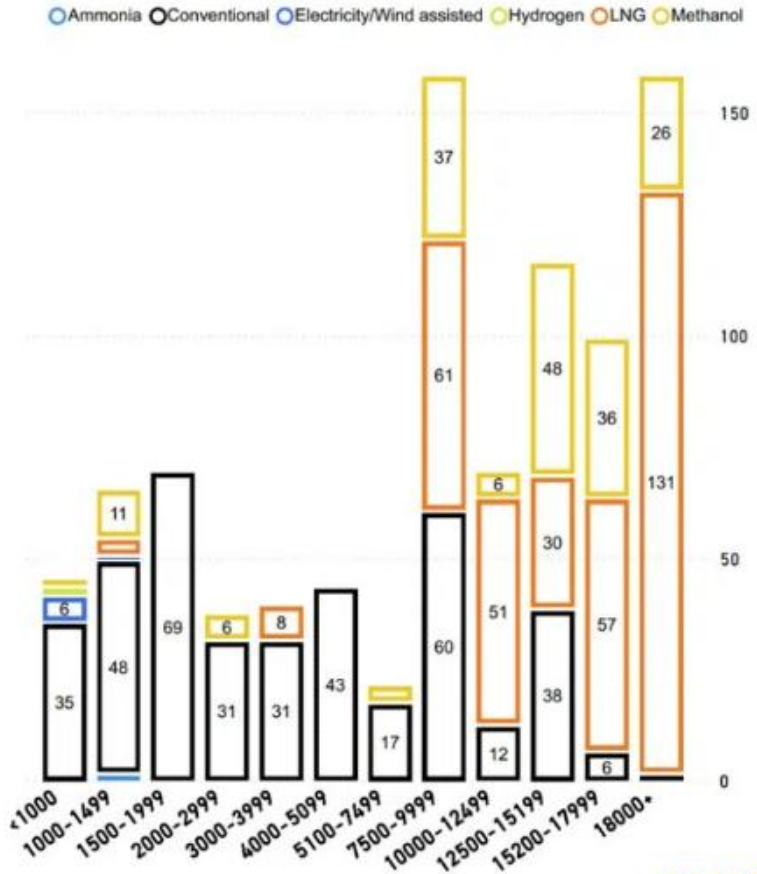
4. Shipbuilding Deliveries

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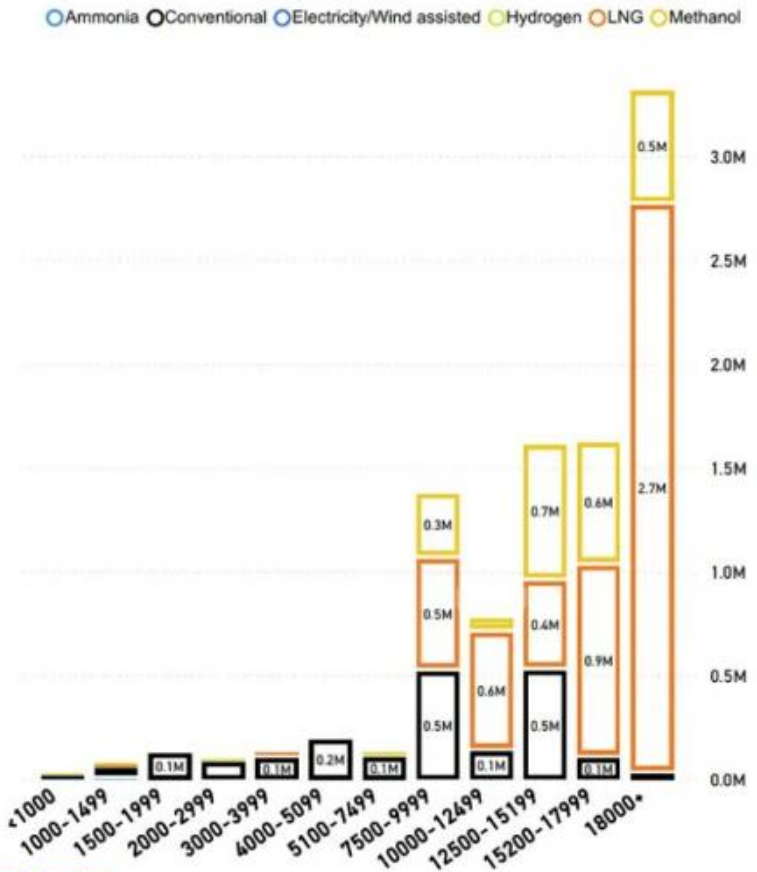
- Shipbuilding is a long-cycle business and time-lag between ordering and delivery can be between 1-4 years depending on the size of orderbook held by shipbuilders.
- Orders must be placed on the basis of an estimate of future demand.

CONTAINER SHIP ORDERBOOK PER SIZE AND PROPULSION

UNITS BREAKDOWN

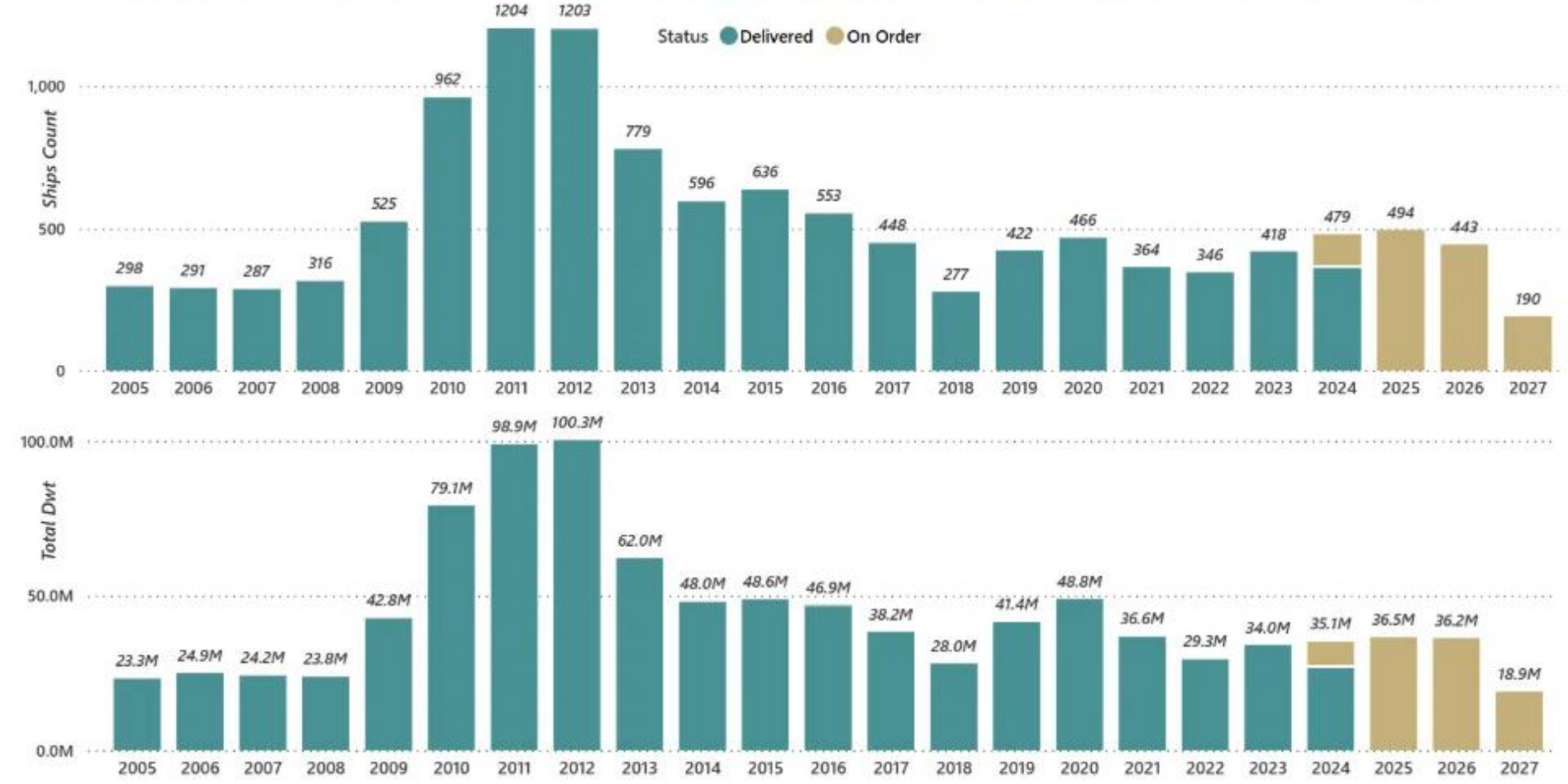


TEU BREAKDOWN

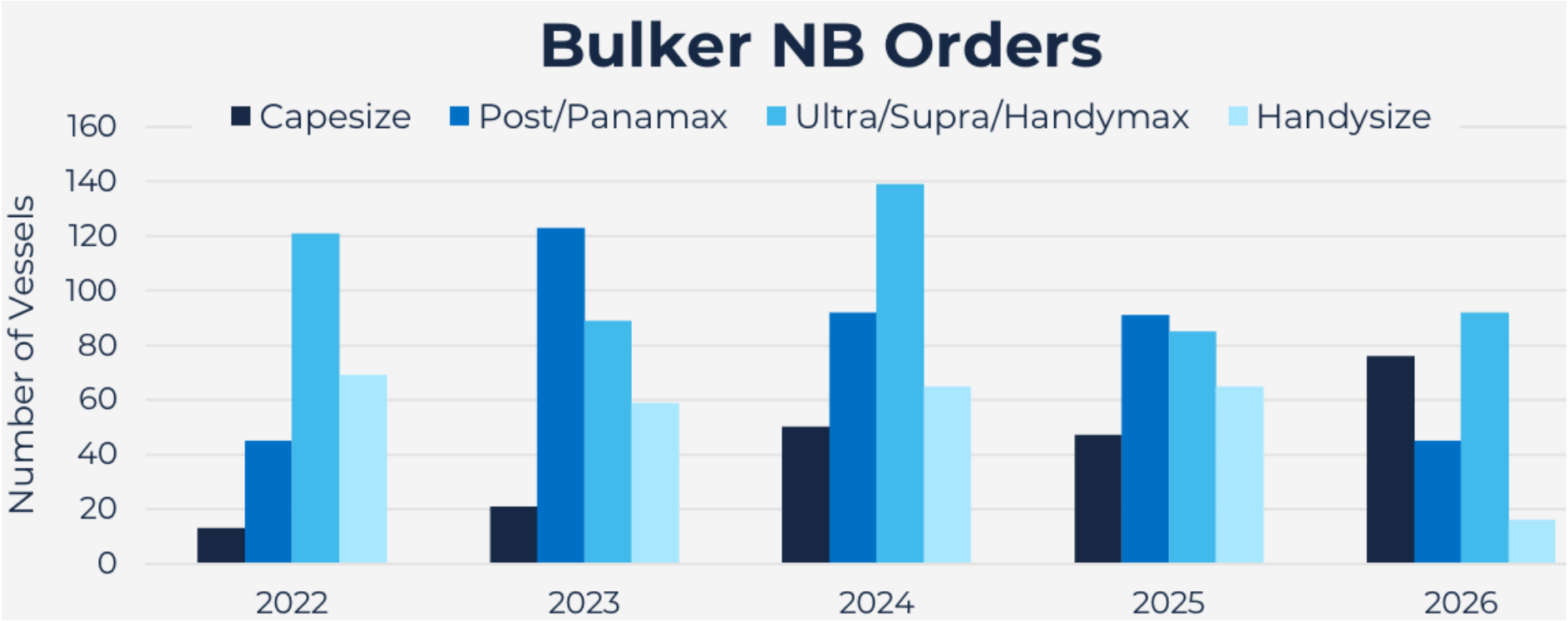


ALPHALINER

DRY BULK DELIVERIES & CURRENT ORDERS

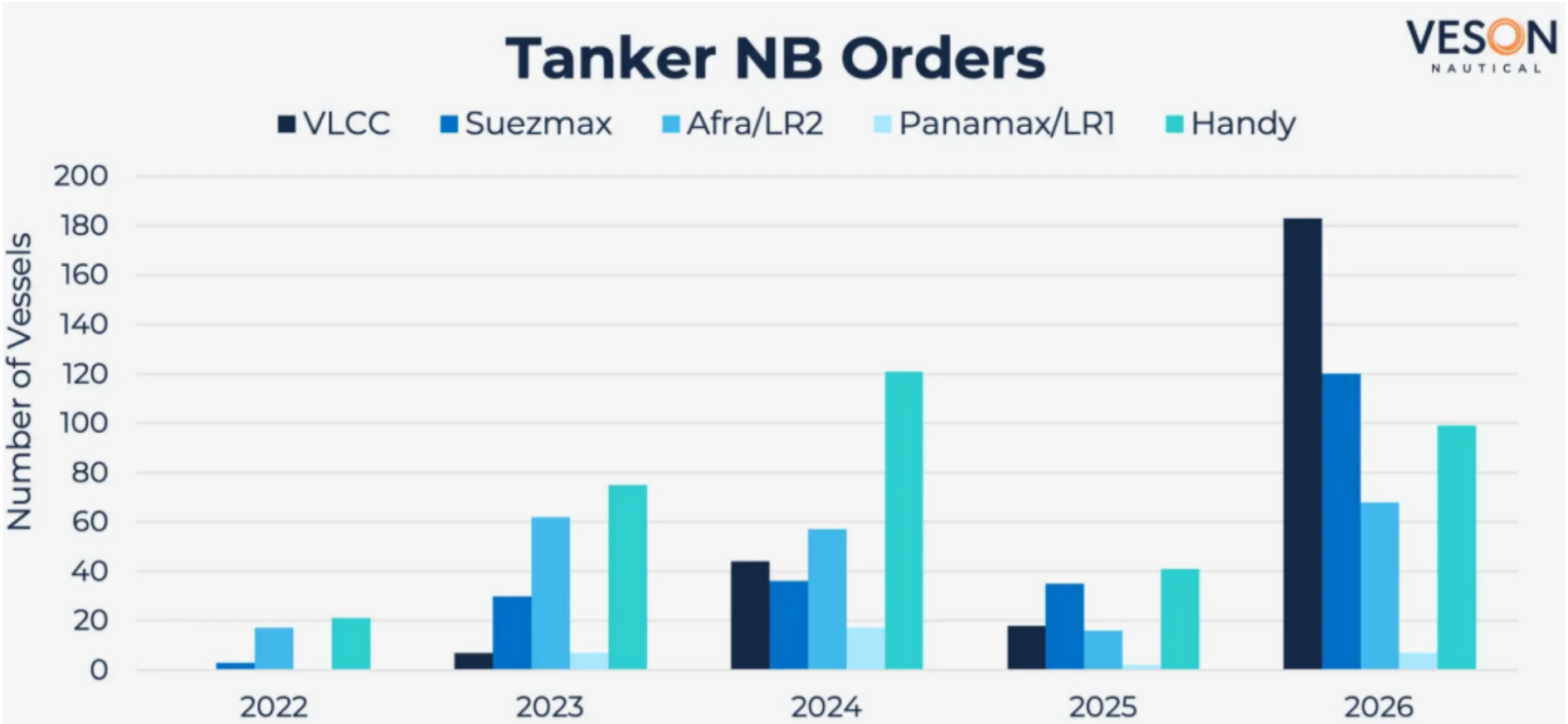


Bulker NB Orders



Tanker NB Orders

VESON NAUTICAL



Top 10 Bulker Buying Companies of 2026 by Total Spend

VESON NAUTICAL

Company	Number of vessels	Total spend (USD M)
Capital Axis Maritime	8	596.0
ICBC Financial Leasing	7	504.3
Maran Dry Management	6	442.8
CMB Financial Leasing	6	375.8
Bank of Comms Financial Leasing	4	254.2
Seanergy Maritime Corp	3	229.7
Cape Shipping SA	3	228.0
Asyad Shipping	5	184.9
JIC Leasing	8	180.5
Mercuria	4	153.1

Top 10 Tanker Buying Companies of 2026 by Total Spend

VESON NAUTICAL

Company	Number of vessels	Total spend (USD M)
Sinokor	73	5,925.0
Industrial Bank Financial Leasing	12	1,151.9
ADNOC Logistics and Services	8	987.0
Bank of Comms Financial Leasing	8	917.8
Frontline	7	875.7
Trafigura Beheer BV	9	778.5
Pan Ocean	10	676.2
CMB Financial Leasing	9	615.0
Torm	15	555.3
BOC Financial Leasing	6	466.8

Race to bigger boxships could resume as carriers weigh 27,000+ teu giants

- *Ships of 27,000+ teu back on the agenda as Alphaliner argues changing economics could revive containership upsizing*
- *Maersk seen as the leading candidate to pioneer a new generation of ultra large vessels as older flagship ships approach replacement age*
- *Ports and safety remain major obstacles, with industry executives warning that infrastructure constraints may limit further growth in vessel size*

09 Sep 2026 | **NEWS**

Torm returns to newbuilds amid soaring secondhand values and record earnings

- *Torm has signed its first direct newbuilding contracts since 2020, ordering six conventional-fuel MR2 product tankers from China's Zhoushan Changhong International Shipyard*
- *The order increases Torm's newbuilding backlog to 12 ships, following six MR vessels secured through resale deals earlier this year*
- *The announcement comes alongside record quarterly results as Q2 revenue more than doubled year on year to $663m, while net profit rose to $338m from $59m*

26 Aug 2026 | **NEWS**

Vehicle carrier newbuild boom continues as Ray Carrier Carriers signs 10-ship order

- *Tonnage provider Ray Car Carriers has ordered 10 pure car and truck carriers from China's Guangzhou Shipyard International, in a deal worth more than $1bn*
- *The 8,200 ceu vessels will be delivered between 2029 and 2031 and are the Isle of Man-based owner's first PCTC order at a Chinese shipyard*

03 Sep 2026 | **NEWS**

Review Question

- Which of the following will decrease fleet productivity?

a. Increasing speed

b. Improving deadweight utilization

c. Increasing non-trading days.

d. Finding commodities for back haul journey.

Answer:



5. Scrapping and Losses

5. Scrapping and Losses

- Rate of growth of merchant fleet depends on the balance between deliveries of new ships and deletions from the fleet in the form of ships scrapped or lost at sea.
- Explaining or predicting the age at which a ship will actually be scrapped is an extremely complex matter.

5. Scrapping and Losses

- Dependent on:
  - i. **Age**
  - ii. **Technical obsolescence**
  - iii. **Scrap prices**
  - iv. **Current earnings (financial position)**
  - v. **Owner's expectations**

5.1 Age

- Primary factor determining tonnage of vessels scrapped.
- As ships get older, the _____ and _____.
- No specific age at which a ship is scrapped, depends typically on its market or technical reason.

5.2 Technical Obsolescence

- May reduce the age at which a particular type of vessel is scrapped because it is superseded by a more efficient ship type.
- Extends to the ship's machinery and gear.
- Affected by _____.

5.3 Scrap Prices

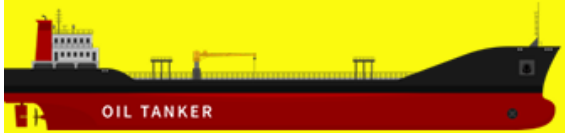
- Decision to scrap also influenced by scrap prices > steel industry.
- Scrap prices fluctuate widely depending upon the state of demand and supply in the steel industry and availability of scrap metal from shipbreaking or demolition of vehicles.
- Period of extensive ship scrapping may even depress prices of scrap metal.

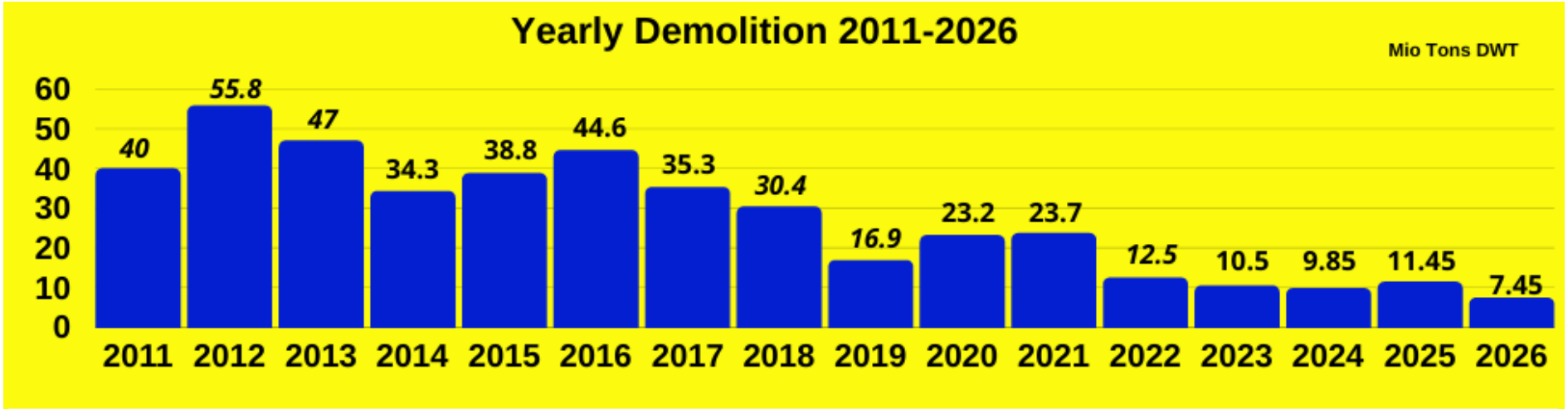
Ship Recycling Market Report

Week 36 (31st August 2026 to 6th September 2026)

WEEKLY TREND: UPWARDS

$US/LT Ldt

	India	Bangladesh	Pakistan	Turkey
	$445	$480	$505	$271
	$460	$515	$520	$281
	$480	$525	$530	$291



HISTORICAL DEMOLITION PRICES*

	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
TANK	482	405	435	425	300	300	440	435	380	410	570	550	520	475	385	490
DRY	456	375	410	400	285	280	425	425	370	390	555	550	520	475	385	490

*Indian End of year prices

5.4 Current Earnings and Owner's Expectations

- Scrapping of a ship is a business decision.
- Depends on owner's expectations of the future operating profitability of the vessel and his financial position.
- During recession, **IF OWNER BELIEVES** there is small chance of freight market boom in near future:
  - unlikely to sell unprofitable ships for scrap;
  - justify incurring a small operating loss



6. Freight Revenue

6. Freight Revenue

How do shipping firms adjust their supply of shipping services?

The supply of shipping facilities cannot expand or contract in the _____ run.

In the _____ run, there is a time lag between entrepreneurs' decision to expand their fleets and the actual time of delivery of new vessels.

Thus, the supply of shipping services tends to be inelastic and incapable of responding instantly to demand and freight rate changes.

6.1 Short Run

It is unlikely to expand or reduce the number of ships in the short run. However, it does not mean that active supply remains the same.

If freight rates are _____ and if carriers anticipate an upward trend in the freight market, active supply will increase as a result of the following developments:

- **Postponement of periodic surveys and maintenance**
- Maximization of the possible **service speed at sea**
- Acceleration of the **processes of loading and unloading** in order to reduce the berthing time
- **Reduction in laid-up tonnages**

6.1 Short Run

On the other hand, if freight rates are _____ and if carriers do not anticipate an improvement in the shipping market, active supply will decrease as a result of the following:

Decrease in the average speed of vessels at sea, i.e. slow steaming.	**Carrying quantities of cargo that are less than the maximum cargo-carrying capacity of vessels**.	**Slow operation** for loading and discharging.	**Laying up of vessels.**

6.1 Short Run

_____ Layup		
usually relatively **short term**.	reduced number of crew on board and some machinery working.	vessel can be reactivated with less cost, time and effort.

6.1 Short Run

_____ Layup		
Usually relatively long **term,** vessel anchored in secure location	all systems shut down with minimum ongoing maintenance	Specialist lay-up crew may be employed.

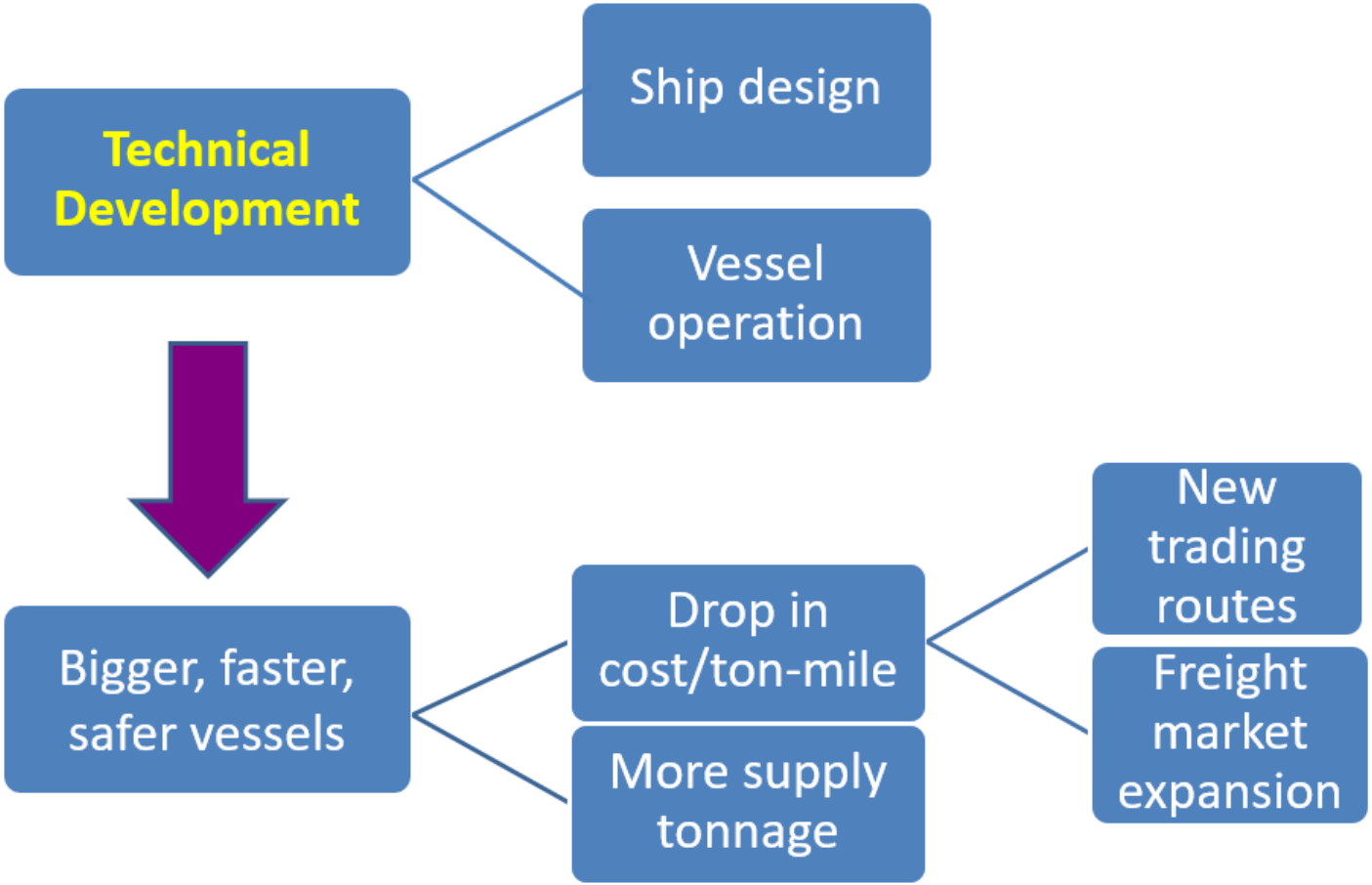
6.2 Long Run

Long run is a term used to denote a period over which full adjustment to change can take place. It refers to the period in which capital stock, such as ships, can be replaced. It also denotes **the time over which shipping firms can enter or leave the shipping industry**.

The magnitude of supply in the long run depends on the following factors:

The **level of demand** for shipping services.	**Carriers' expectations** regarding the freight rates of the shipping market.	**Technical development** as technical progress and innovations in shipbuilding can contribute to more economically efficient production of sea transport services.

6.2 Long Run


Technical Development
Ship design
Vessel operation
Bigger, faster, safer vessels
Drop in cost/ton-mile
More supply tonnage
New trading routes
Freight market expansion

Review Question

When freight rate is low, _____.

a. In the short run, shipowner can increase speed and in the long run, the shipowner can scrap ships.
b. In the short run, shipowner can reduce speed and in the long run, shipowner can order ships.
c. In the short run, shipowner can reduce speed and in the long run, shipowner can scrap ships.
d. In the short run, shipowner can increase speed and in the long run, shipowner can order ships.

Answer:

Summary

Summary

- The rules of the shipping market game are set by the economic relationships which create freight rates.
- As demand for ships changes rapidly, but supply is slow and ponderous, freight cycles are generally irregular.
- In order to have a clear picture and/or forecast of the world shipping supply, have to view the factors collectively, rather than individually.